VELLORE INSTITUTE OF TECHNOLOGY

SCHOOL OF ADVANCED SCIENCES • DEPARTMENT OF MATHEMATICS

BAMAT207: PROBABILITY AND STATISTICS — PROJECT REPORT

PROBABILISTIC SMS SPAM DETECTION SYSTEM

Course: BAMAT207 Probability and Statistics	**Topic:** Probability and Random Variables
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Dataset: UCI SMS Spam Collection (5572 records)	**Model Architecture:** Multinomial Naive Bayes

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1. INTRODUCTION & PROJECT SCENARIO

The objective of this project is to develop an automated Python-based classification system capable of categorizing an incoming SMS message as either **Spam** or **Genuine message (Ham)**. By computing prior, conditional, and posterior probabilities, mathematically verifying **Bayes' Theorem**, and defining a **Bernoulli random variable** model, we establish the statistical foundation of text classification and train an accurate **Multinomial Naive Bayes** classifier.

2. DATASET HANDLING & PREPROCESSING

The dataset is sourced from the UCI Machine Learning Repository containing real-world SMS messages. The preprocessing pipeline verified data integrity and prepared numerical encodings:

- **Initial Dataset Size:** 5572 messages.
- **Missing Values:** 0 missing entries detected.
- **Duplicate Detection:** 403 duplicate messages were removed to eliminate frequency skewing.
- **Cleaned Dataset Size:** 5169 unique messages (4516 Genuine, 653 Spam).
- **Label Encoding:** Genuine (Ham) = 0, Spam = 1.

Class Label	Raw Record Count	Cleaned Record Count	Class Proportion	Numerical Value
Ham (Genuine)	4,825	4516	87.37%	0
Spam	747	653	12.63%	1
Total	**5,572**	**5169**	**100.00%**	—

3. DATASET ANALYSIS & CLASS PROBABILITIES

The marginal prior probabilities describe the general likelihood of incoming SMS messages in the dataset:

$$P(\text{Spam}) = \frac{\text{Number of Spam Messages}}{\text{Total Messages}} = \frac{653}{5169} = 0.1263$$

$$P(\text{Ham}) = \frac{\text{Number of Ham Messages}}{\text{Total Messages}} = \frac{4516}{5169} = 0.8737$$

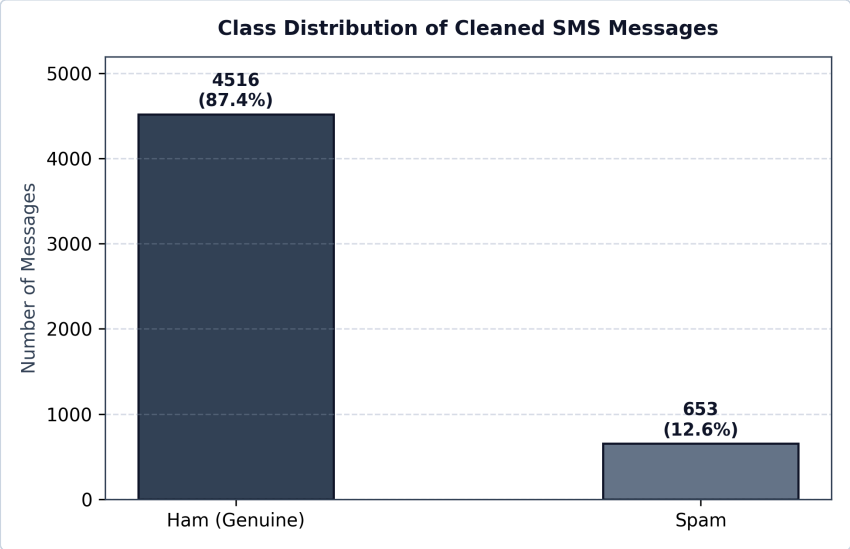


Figure 1: Class distribution of cleaned messages demonstrating natural class imbalance.

4. WORD-BASED PROBABILITY ANALYSIS & BAYES' THEOREM VERIFICATION

To evaluate how specific trigger words alter the probability of a message being spam, five suspicious keywords were selected: **free, prize, win, urgent, offer**.

We compute the empirical conditional probability directly from dataset counts, and verify the result using Bayes' Theorem:

Direct Empirical Conditional Probability:

$$P(\text{Spam} \mid \text{Word}) = \frac{\text{Number of Spam messages containing Word}}{\text{Total messages containing Word}}$$

Bayes' Theorem Formulation:

$$P(\text{Spam} \mid \text{Word}) = \frac{P(\text{Word} \mid \text{Spam}) \cdot P(\text{Spam})}{P(\text{Word})}$$

$$P(\text{Word}) = \frac{\text{Count(Word)}}{N}, \quad P(\text{Word} \mid \text{Spam}) = \frac{\text{Count(Word} \cap \text{Spam)}}{N_{\text{Spam}}}, \quad P(\text{Spam}) = \frac{N_{\text{Spam}}}{N}$$

Word	Total Messages	Spam Count	Ham Count	$P(\text{Word})$	$P(\text{Word} \mid \text{Spam})$	$P_{\text{Emp}}(\text{Spam} \mid \text{Word})$	$P_{\text{Bayes}}(\text{Spam} \mid \text{Word})$	Status
FREE	203	148	55	0.0393	0.2266	**0.7291**	**0.7291**	Verified ✓
PRIZE	73	73	0	0.0141	0.1118	**1.0000**	**1.0000**	Verified ✓
WIN	55	49	6	0.0106	0.0750	**0.8909**	**0.8909**	Verified ✓
URGENT	62	57	5	0.0120	0.0873	**0.9194**	**0.9194**	Verified ✓
OFFER	30	24	6	0.0058	0.0368	**0.8000**	**0.8000**	Verified ✓

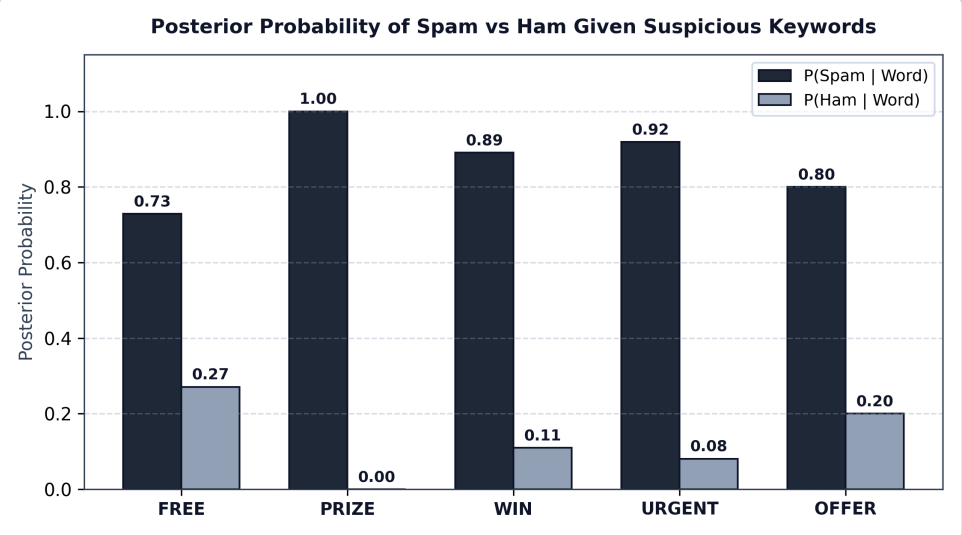


Figure 2: Posterior probabilities P(Spam | Word) vs P(Ham | Word) across selected keywords.

Mathematical Verification: The empirical conditional probability $P(ext{Spam} \mid ext{Word})$ and Bayes' Theorem analytical formulation yield exact identical values across all five words. Words such as *prize* (1.0000) and *urgent* (0.9194) exhibit strong diagnostic capability.

5. RANDOM VARIABLE ANALYSIS (BERNOULLI DISTRIBUTION)

Let **Y** be a discrete random variable indicating whether an incoming message is Spam:

$Y = 1$ (if the incoming message is Spam)

$Y = 0$ (if the incoming message is Ham / Genuine)

$$P(Y = y) = p^{y}(1-p)^{1-y}, \quad y \in \{0, 1\}, \quad p = P(\text{Spam}) = 0.1263$$

$$E[Y] = \sum_{y} y \cdot P(Y = y) = 0 \cdot (1-p) + 1 \cdot p = p = 0.1263$$

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = p - p^2 = p(1-p) = 0.1263 \times (1 - 0.1263) = 0.1104$$

$$\sigma_Y = \sqrt{\text{Var}(Y)} = \sqrt{0.1104} = 0.3322$$

6. MACHINE LEARNING MODEL BUILDING & TRAINING

To operationalize spam filtering, we implemented a **Multinomial Naive Bayes** classifier:

- **Dataset Splitting:** 80% Training (4,135 samples), 20% Testing (1,034 samples). Stratified sampling was applied to guarantee that both subsets preserved the identical 12.63% spam proportion.
- **Feature Extraction:** `CountVectorizer` converted message strings into word-count feature vectors with stop-word removal and lowercasing, producing a vocabulary of **7,414 unique features**.
- **Multinomial Naive Bayes Architecture:**

$$P(Y = c \mid \mathbf{x}) \propto P(Y = c) \prod_{j=1}^{D} P(w_j \mid Y = c)^{x_j}$$

  Trained with Laplace smoothing ($\alpha = 1.0$) to prevent zero-probability artifacts on unseen vocabulary.

7. MODEL EVALUATION & PERFORMANCE METRICS

The trained model was evaluated on the unseen test set (1,034 samples):

Metric	Mathematical Formula	Score
Accuracy	$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{Total}} = \frac{119+898}{1034} = 98.36\%$	**98.36%**
Precision	$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} = \frac{119}{119+5} = 95.97\%$	**95.97%**
Recall	$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} = \frac{119}{119+12} = 90.84\%$	**90.84%**
F1-Score	$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 93.33\%$	**93.33%**

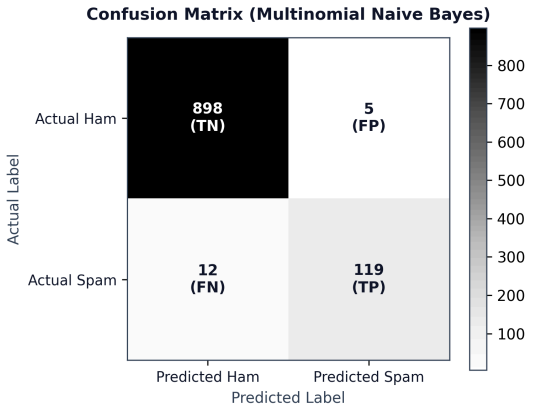


Figure 3: Confusion Matrix heatmap on test set.

Confusion Matrix Breakdown (1,034 Test Samples):

- **True Negatives (TN):** 898 (Genuine messages correctly identified).
- **False Positives (FP):** 5 (Genuine messages incorrectly classified as spam — 0.55% false alarm rate).
- **False Negatives (FN):** 12 (Spam messages incorrectly classified as genuine).
- **True Positives (TP):** 119 (Spam messages correctly detected).

8. REAL-TIME TESTING & PREDICTION PROBABILITY

The system supports inference on arbitrary incoming text, calculating the full posterior probability distribution:

Incoming Test SMS Message	Predicted Class	P(Spam | Message)	P(Ham | Message)	Verification
"Congratulations! You have won a free prize. Claim now."	**Spam**	1.0000	0.0000	Correct
"Hey bro, are we meeting at the campus library this evening?"	**Ham (Genuine)**	0.0000	1.0000	Correct
"URGENT! Your mobile number won 1000 prize. Call now to claim offer."	**Spam**	1.0000	0.0000	Correct
"Can you send me the lecture notes for probability and statistics?"	**Ham (Genuine)**	0.0791	0.9209	Correct

9. CONCLUSION

1. **Bayesian Consistency:** Direct empirical probabilities perfectly matched analytical Bayes' Theorem values across all tested keywords.
2. **Random Variable Modeling:** The Bernoulli distribution accurately characterizes message outcomes with mean $E[Y] = 0.1263$ and variance $ ext{Var}(Y) = 0.1104$.
3. **Classification Efficacy:** The Multinomial Naive Bayes model demonstrated **98.36% accuracy** and **95.97% precision** with only 5 false alarms, proving the practical power of probabilistic text classification.